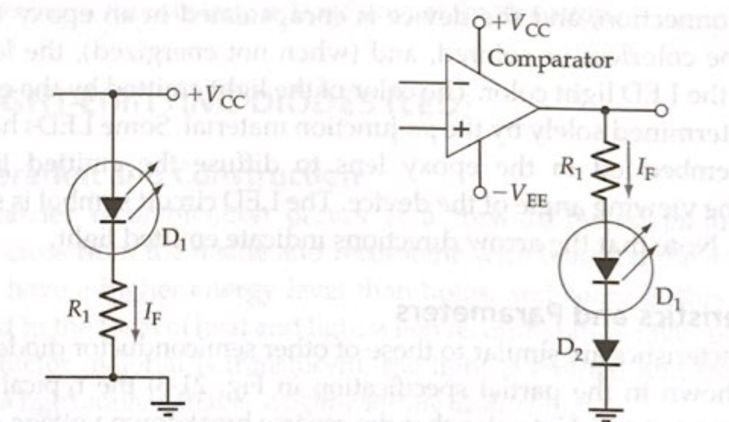
Typical LED Specification	Min	Typ	Max
Luminous intensity (I_V at 20 mA)	4 mcd	8 mcd	
Forward voltage (V_F)	1.4 V	1.6 V	2.0 V
Reverse breakdown voltage (V_{FBR})	3 V	10 V	
Peak forward current ($I_{F(max)}$)		90 mA	
Average forward current ($I_{F(av)}$)		20 mA	
Power dissipation (P_D)		100 mW	
Response speed (τ_S)		90 ns	
Peak wavelength (λ_p)		660 nm	

Figure 21-3 Partial specification for a typical light-emitting diode.

LED Circuits

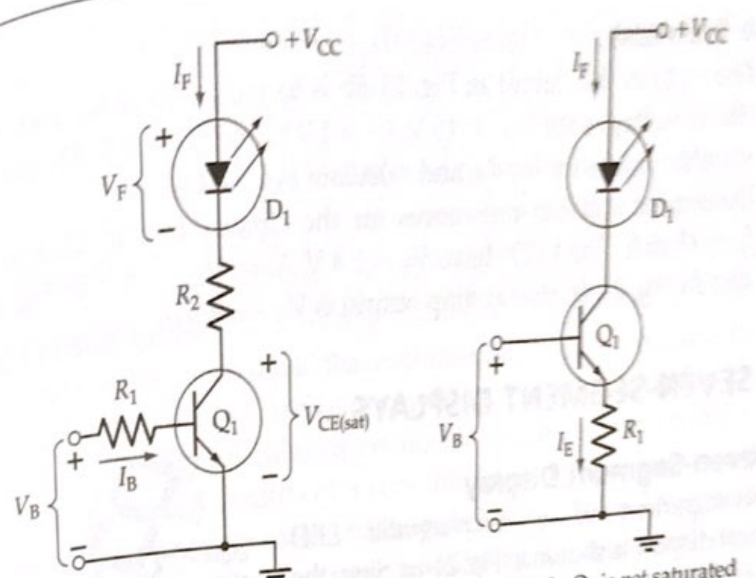
As explained, an LED is a semiconductor diode that emits light when a forward current is passed through the device. A single LED might be employed simply as a supply voltage *on/off* indicator, as illustrated in Fig. 21-4a. A series-connected resistor (R_1) must be included to limit the current to the desired level. Figure 21-4b shows an LED connected at the output of a comparator to indicate a *high* output voltage. As well as the current-limiting resistor (R_1), an ordinary semiconductor diode (D_1) is connected in series with the LED to protect it from an excessive reverse voltage when the comparator output is negative.



(a) LED as *on/off* indicator (b) LED as *high* output voltage indicator

Figure 21-4 Light-emitting diode indicator circuits.

LEDs are often controlled by a BJT, as illustrated in Figs 21-5a and b. Transistor Q_1 in Fig. 21-5a is switched into saturation by the input voltage (V_B). Resistor R_1 limits the transistor base current, and R_2 limits the LED current. In Fig. 21-5b, the emitter resistor (R_1) limits the LED current to $(V_B - V_{BE})/R_1$.



(a) R_2 controls I_F, Q_1 is saturated (b) R_1 controls I_F, Q_1 is *not* saturated

Figure 21-5 BJT control circuits for light-emitting diodes.

Example 21-3

The LED in Fig. 21-5a is to have a forward current of approximately 10 mA. The circuit voltages are $V_{CC} = 9$ V, $V_F = 1.6$ V, and $V_B = 7$ V; and Q_1 has $h_{FE(min)} = 100$. Calculate suitable resistance values for R_1 and R_2.

Solution

$$R_2 = \frac{V_{CC} - V_F - V_{CE(sat)}}{I_C} = \frac{9\text{ V} - 1.6\text{ V} - 0.2\text{ V}}{10\text{ mA}}$$

$$= 720\ \Omega \text{ (use } 680\ \Omega \text{ standard value)}$$

I_C becomes

$$I_C = \frac{V_{CC} - V_F - V_{CE(sat)}}{R_2} = \frac{9\text{ V} - 1.6\text{ V} - 0.2\text{ V}}{680\ \Omega}$$

$$= 10.6\text{ mA}$$

$$I_B = \frac{I_C}{h_{FE(min)}} = \frac{10.6\text{ mA}}{100}$$

$$= 106\ \mu\text{A}$$

$$R_B = \frac{V_B - V_{BE}}{I_B} = \frac{7\text{ V} - 0.7\text{ V}}{106\ \mu\text{A}}$$

$$= 59\text{ k}\Omega \text{ (use } 56\text{ k}\Omega \text{ standard value)}$$